UNITED STATES
NUCLEAR REGULATORY COMMISSION
WASHINGTON, D.C. 20555-0001

August 4, 2020

Mr. Gale Voyles
Nuclear Quality Assurance Manager
Valcor Engineering Corporation
2 Lawrence Road
Springfield, NJ 07081

SUBJECT: NUCLEAR REGULATORY COMMISSION VENDOR INSPECTION REPORT OF VALCOR ENGINEERING CORPORATION, NO. 99900728/2020-201

Dear Mr. Voyles:

From July 13 through July 17, 2020, the U.S. Nuclear Regulatory Commission (NRC) staff conducted a limited-scope virtual inspection of Valcor Engineering Corporation (hereafter referred to as Valcor), located in Springfield, NJ. The purpose of this limited-scope virtual inspection was to verify that Valcor's corrective actions initiated to address the findings identified during the NRC inspection performed in October 2017 were adequately implemented and met the applicable requirements of Criterion XVI, "Corrective Action," of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities."

This technically-focused virtual inspection specifically evaluated Valcor's implementation and closure of the corrective actions taken in response to Notice of Nonconformance (NON) 99900728/2017-201-01 and NON 99900728/2017-201-03, documented in the NRC's inspection report No. 99900728/2017-201, dated November 16, 2017 (Agencywide Documents Access and Management System Accession (ADAMS) No. ML17311A267). During the October 2017 inspection, the NRC inspection team inspected records associated with inspections, tests, analyses, and acceptance criteria (ITAAC) from Revision 19 of the certified AP1000 Design Control Document. Specifically, these activities were associated with ITAAC Nos. 2.1.02.07a.i, 2.2.05.02a, 2.2.05.03a, and 2.2.05.04a, for Vogtle Electric Generating Plant, Units 3 and 4. The NRC inspection team determined that NON 99900728/2017-201-01 was material to the ITAAC acceptance criteria for ITAAC No. 2.1.02.07a.i. During this inspection, the NRC inspection team determined that Valcor's corrective actions were adequate to address NON 99900728/2017-201-01. Based on the review of the corrective actions, the NRC inspection team closed NON 99900728/2017-201-01. The enclosed report presents the results of the inspection. This NRC inspection report does not constitute NRC endorsement of Valcor's overall quality assurance (QA) program.

This inspection was performed virtually due to the ongoing worldwide pandemic associated with the novel Coronavirus. Due to the challenges and limitations associated with performing a virtual inspection, the NRC inspection team did not close NON 99900728/2017-201-02, consequently, this NON will remain open. The NRC may review the implementation of your corrective actions for this NON during a future on-site inspection to determine whether full compliance has been achieved and maintained.

Based on the results of this virtual inspection, the NRC inspection team found that the implementation of your QA program with regards to the corrective actions associated with NON 9900728/2017-201-01 and NON 99900728/2017-201-03 met the applicable requirements of Criterion XVI of Appendix B to 10 CFR Part 50. No findings of significance were identified.

In accordance with 10 CFR 2.390, “Public inspections, exemptions, requests for withholding,” of the NRC's “Rules of Practice,” a copy of this letter and its enclosures will be made available electronically for public inspection in the NRC’s Public Document Room and in ADAMS, accessible from the NRC’s public web site at http://www.nrc.gov/reading-rm/adams.html.

Sincerely,

Kerri A. Kavanagh, Chief **/RA/**
Quality Assurance and Vendor Inspection Branch
Division of Reactor Oversight
Office of Nuclear Reactor Regulation

Docket No.: 99900728

EPID No.: I-2020-201-0040

Enclosures: Inspection Report No. 99900728/2020-201 and Attachment

EXECUTIVE SUMMARY

Valcor Engineering Corporation
99900728/2020-201

From July 13 through July 17, 2020, the U.S. Nuclear Regulatory Commission (NRC) staff conducted a limited-scope virtual inspection of Valcor Engineering Corporation (hereafter referred to as Valcor), located in Springfield, NJ. The purpose of this limited-scope virtual inspection was to evaluate Valcor's corrective actions initiated to address the findings identified during the NRC inspection performed in October 2017 to ensure they were adequately implemented and met the applicable requirements of Criterion XVI, "Corrective Action," of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities."

This technically-focused virtual inspection specifically evaluated Valcor's implementation and closure of the corrective actions taken in response to Notice of Nonconformance (NON) 99900728/2017-201-01 and NON 99900728/2017-201-03, documented in the NRC's inspection report No. 99900728/2017-201, dated November 16, 2017 (Agencywide Documents Access and Management System Accession No. ML17311A267).

These regulations served as the bases for the NRC inspection:

- Appendix B to 10 CFR Part 50
- 10 CFR Part 21

During the course of this virtual inspection, the NRC inspection team implemented portions of Inspection Procedure (IP) 36100, "Inspection of 10 CFR Part 21 and Programs for Reporting Defects and Noncompliance," dated May 16, 2019; IP 43002, "Routine Inspections of Nuclear Vendors," dated January 27, 2017; and IP 43004, "Inspection of Commercial-Grade Dedication Programs," dated January 27, 2017.

This inspection was performed virtually due to the ongoing worldwide pandemic associated with the novel Coronavirus. Due to the challenges and limitations associated with performing a virtual inspection, the NRC inspection team was not able to close NON 99900728/2017-201-02, consequently, this NON will remain open. In addition, the NRC inspection team could not perform an assessment of Valcor's Safety Conscious Work Environment (SCWE) due to the inspection being performed virtually. The NRC may review the implementation of the corrective actions opened for this NON to determine whether full compliance has been achieved and maintained and may assess Valcor's SCWE during a future on-site inspection.

The results of the inspection are summarized below.

Corrective Action

The NRC inspection team reviewed Valcor's policies and implementing procedures that govern the implementation of its corrective action program to determine compliance with the requirements of Criterion XVI of Appendix B to 10 CFR Part 50. Specifically, the NRC inspection team reviewed the implementation and closure of the corrective actions taken by Valcor in response to NON 99900728/2017-201-01 and NON 99900728/2017-201-03, documented in the October 2017 inspection report. NON 99900728/2017-201-01 is material to

the acceptance criteria of Inspection, Tests, Analyses, and Acceptance Criteria (ITAAC) No. 2.1.02.07a.i for Vogtle Electric Generating Plant, Units 3 and 4, from Revision 19 of the certified AP1000 Design Control Document. The NON stated, in part, that as part of the design validation process, Valcor did not perform sufficient testing or analysis of the AP1000 solenoid valves to verify that the voltage controllers would be capable of providing an acceptable output to the valves, including those valves defined as being in a harsh environment resulting from a design basis accident. NON 99900728/2017-201-03 stated, in part, that Valcor calibrated a durometer (serial No. 120978/KRW006) using a rubber test block kit (serial No. 120978/WDT004), contrary to the requirements of the American Society for Testing and Materials standard No. D2240, "Standard Test Method for Rubber Property - Durometer Hardness," 2015 Edition.

The NRC inspection team reviewed the documentation that provided objective evidence that all of the corrective actions were completed and adequately implemented. Based on this review and interviews with Valcor's staff responsible for implementing the corrective actions, the NRC inspection team closed 99900728/2017-201-01 and NON 99900728/2017-201-03. No findings of significance were identified.

REPORT DETAILS

1. Corrective Action

a. Inspection Scope

The Nuclear Regulatory Commission (NRC) inspection team reviewed Valcor Engineering Corporation's (hereafter referred to as Valcor) policies and implementing procedures that govern the implementation of its corrective action program to verify compliance with the regulatory requirements of Criterion XVI, "Corrective Action," of Appendix B, "Quality Assurance Program Criteria for Nuclear Power Plants and Fuel Reprocessing Plants," to Title 10 of the *Code of Federal Regulations* (10 CFR) Part 50, "Domestic Licensing of Production and Utilization Facilities." Specifically, the NRC inspection team reviewed the implementation and closure of the corrective actions taken in response to Notice of Nonconformance (NON) 99900728/2017-201-01 and NON 99900728/2017-201-03, documented in the NRC's inspection report no. 99900728/2017-201, dated November 16, 2017 (Agencywide Documents Access and Management System (ADAMS) No. ML17311A267).

The NRC inspection team also discussed the corrective action program with Valcor's management and technical staff. The attachment to this inspection report lists the documents reviewed by the NRC inspection team.

b. Observations and Findings

b.1 Corrective Action Associated with NON 99900728/2017-201-01

Following the October 2017 inspection of Valcor, the NRC issued NON 99900728/2017-201-01 for Valcor's failure to fully verify the adequacy of the design of the solenoid valves' voltage controllers. Specifically, as part of the design validation process, Valcor did not perform sufficient testing or analysis to verify that the voltage controllers would be capable of providing an acceptable output to the valves, including those valves defined as being in a harsh environment resulting from a design basis accident. The NRC staff determined that NON 99900728/2017-201-01 is material to the acceptance criteria of Inspection, Tests, Analyses, and Acceptance Criteria (ITAAC) No. 2.1.02.07a.i for Vogtle Electric Generating Plant, Units 3 and 4, from Revision 19 of the certified AP1000 Design Control Document.

In its response dated December 15, 2017 (ADAMS No. ML17355A114), Valcor stated that it had initiated Corrective Action (CA) No. 2017-41 to address this issue. The response stated that following the NRC inspection, Valcor created Engineering Repot (ER)-045, "Engineering Report for Westinghouse AP1000 Voltage Control Box P/N S1140-23-23 Output Voltage Design Analysis," Revision A, dated December 14, 2017. This report provides the calculated values proving the voltage controllers are capable of providing an acceptable output to the solenoid valves, including those solenoid valves defined as being in a harsh environment resulting from a design basis accident. The analysis performed within this report indicated that the voltage controllers have sufficient margin to meet all of the required operation conditions.

The response also stated that additional testing would be performed to ensure that every production unit output a voltage adequate to initially pull-in the solenoid valves and remains energized after the drop-down voltage is achieved while simulating the worst case hot and cold coil resistance loads. Although not stated in the response, Valcor revised Engineering Test Procedure No. S1477, "Acceptance and Burn-In Test Procedures for both AC and DC VCBs," to include the different resistor values and the requirements for the voltage controller output holding voltage span test.

In addition, the response stated that the test report (TR) for the voltage controller, TR 1140-23-23, "Acceptance Test Report for the DC 1E Voltage Control Box," would be revised to add testing necessary to ascertain the initial output of the voltage controllers when minimum, nominal, and maximum voltage is applied to the voltage controllers. Every voltage controller would be tested with a resistive load cell representing the worst case hot and cold coil resistance loads to verify its operability.

Furthermore, the response stated that commercial-grade dedication plan no. S1140-23-23Q, "Commercial-Grade Item Dedication Plan," would be developed to identify the set of critical characteristics of the voltage controllers.

The NRC inspection team reviewed the documentation that provided the objective evidence for the completion of the corrective actions, including the review of CA-2017-032. The NRC inspection team confirmed that ER-045 and TR 1140-23-23 were revised as discussed above and determined that Valcor adequately verified through analytical calculations and qualification testing that the voltage controllers are capable of providing the acceptable minimum output voltage required to pull-in the solenoid valve, including those solenoid valves defined as being in a harsh environment resulting from a design basis condition and that they have sufficient margin to meet all their required operating conditions. In addition, the NRC inspection team confirmed that the commercial-grade dedication plan no. S1140-23-23Q was developed and included the appropriate critical characteristics.

The NRC inspection team determined that Valcor's corrective actions were adequate to address the NON. Based on the review of the corrective actions, the NRC inspection team closed NON 99900728/2017-201-01. No findings of significance were identified.

b.2 Corrective Action Associated with NON 99900728/2017-201-02

Due to the challenges and limitations associated with performing a virtual inspection, the NRC inspection team did not close NON 99900728/2017-201-02, consequently, this NON was discussed with Valcor's staff and will remain open. The NRC may review the implementation of the corrective actions for this NON during a future on-site inspection to determine whether full compliance has been achieved and maintained.

b.3 Corrective Action Associated with NON 99900728/2017-201-03

Following the October 2017 inspection of Valcor, the NRC issued NON 99900728/2017-201-03 for Valcor's failure to assure that measuring and test devices used in activities affecting quality were properly calibrated. Specifically, Valcor calibrated a durometer (serial No. 120978/KRW006) using a rubber test block kit (serial No. 120978/WDT004), contrary to the requirements of the American Society for Testing and Materials (ASTM) standard No. D2240, "Standard Test Method for Rubber Property - Durometer Hardness," 2015 Edition. Note 2 in Section 10, "Report," of ASTM D2240 states, in part, that "periodic checking of the operation and state of durometer calibration using commercially available rubber test blocks (refer to 7.8), specifically designed for this purpose, is recommended." Subsection 7.8 of Section 7, "Calibration," of ASTM D2240, states, in part, that "the rubber reference block(s) provided for verifying durometer operation and state of calibration are not to be relied upon as calibration standards."

In its response dated December 15, 2017, Valcor stated that the durometer (serial No. 120978/KRW006) was removed from service on October 12, 2017 and sent to an approved calibration facility for calibration in accordance with ASTM D2240. The response stated that the equipment control and calibration card that defines the method of calibration was revised to reflect the correct method of the durometer tester calibration. On October 26, 2017, Valcor was verbally notified by the calibration supplier that the durometer tester was found to be out of calibration and beyond repair. Valcor received formal report of calibration on October 30, 2017 and initiated Material Rejection Report (MRR) No. 853AB to document the failed calibration condition. Valcor also initiated CA-2017-032 to address this issue. On November 7, 2017, a newly purchased and replacement durometer hardness REX with serial No. KRW006, calibrated in accordance with ASTM D2240, was placed into service.

The response also stated that on November 27, 2017, per disposition by the Material Review Board and as documented on Valcor's MRR No. 853AB, the durometer with serial No. 120978/KRW006 was scrapped and Valcor initiated a 10 CFR Part 21 evaluation in accordance with Valcor's internal procedure S2110, "10CFR21 Defects and Non-Compliance Reporting Procedure," Revision K, dated April 7, 2017. The NRC inspection team confirmed that this evaluation did not result in a reportable condition.

In addition, the response stated that standard S1618, "Calibration Procedure Control Requirement," Revision G, dated April 7, 2017, would be revised to include a requirement for all new specialty gages to be entered into Valcor's calibration system to have calibration method reviewed and approved by Quality Assurance. Furthermore, the response stated that a training session would be held after the document release to make sure that all calibration technicians were aware of the new requirement.

The NRC inspection team reviewed the documentation that provided the objective evidence for the completion of the corrective actions, including the review of CA-2017-032 and MRR No. 853AB. The NRC inspection team confirmed that S1618 was revised to include the requirement as stated above. In

addition, the NRC inspection team reviewed the training log and confirmed that the calibration technicians were trained on the update to the procedure. Furthermore, the NRC inspection team reviewed the 10 CFR Part 21 evaluation and supporting documentation and confirmed that the extent of condition and associated analyses were adequately performed.

The NRC inspection team determined that Valcor's corrective actions were adequate to address the NON. Based on the review of the corrective actions, the NRC inspection team closed NON 99900728/2017-201-03. No findings of significance were identified.

c. Conclusion

The NRC inspection team concluded that Valcor has adequately implemented its program for corrective actions associated with NON 99900728/2017-201-01 and NON 99900728/2017-201-03, described in NRC inspection report No. 99900728/2017-201, dated November 16, 2017. Based on the limited sample of documents reviewed, the NRC inspection team determined that Valcor has taken adequate corrective actions to resolve the non-conforming conditions identified in the subject NONs. No findings of significance were identified.

2. Entrance and Exit Meetings

On July 13, 2020, the NRC inspection team discussed the scope of the inspection with John Trezza, Valcor's Chief Executive Officer and President, and other members of Valcor's management and technical staff. On July 17, 2020, the NRC inspection team presented the inspection results and observations during an exit meeting with Theresa Conway, Valcor's Executive Vice President of Controlling and Finance, and other members of Valcor's management and technical staff. The attachment to this report lists the attendees of the entrance and exit meetings, as well as those individuals whom the NRC inspection team interviewed.